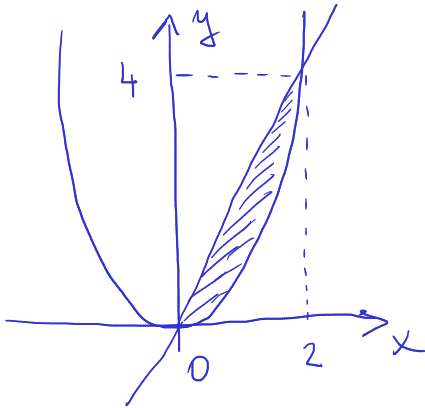


BONUS 4 - ŘEŠENÍ!

①



$$a) \int_0^2 \left[\int_{x^2}^{2x} (x+2y) dy \right] dx$$

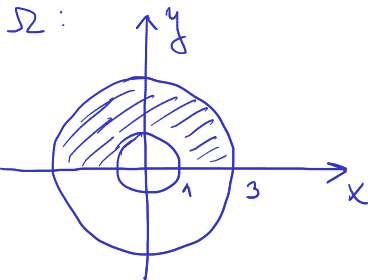
$$\int_0^4 \left[\int_{\frac{y}{2}}^{\sqrt{y}} (x+2y) dx \right] dy$$

$$b) \int_0^2 \left[\int_{x^2}^{2x} (x+2y) dy \right] dx = \int_0^2 \left[xy + y^2 \right]_{x^2}^{2x} dx$$

$$= \int_0^2 (2x^2 + 4x^2 - x^3 - x^4) dx = \int_0^2 (6x^2 - x^3 - x^4) dx$$

$$= \left[2x^3 - \frac{x^4}{4} - \frac{x^5}{5} \right]_0^2 = 16 - 4 - \frac{32}{5} = \underline{\underline{\frac{28}{5}}}$$

②



$$0 \leq \varphi \leq \pi$$

$$1 \leq r \leq 3$$

$$x = r \cdot \cos \varphi$$

$$y = r \cdot \sin \varphi$$

$$x^2 + y^2 = 1$$

$$\iint_{\Omega} \frac{3y+1}{\sqrt{x^2+y^2}} dx dy = \int_0^{\pi} \left[\int_1^3 \frac{3r \cdot \sin \varphi + 1}{\sqrt{r^2}} \cdot r dr \right] d\varphi$$

$$= \int_0^{\pi} \left[\int_1^3 (3r \cdot \sin \varphi + 1) dr \right] d\varphi$$
